

DS lab 11 b.c - Code::Blocks 25.03

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```
1  #include <stdio.h>
2  #include <stdlib.h>
3  #define MAX 10
4  struct Node
5  {
6      int vertex;
7      struct Node *next;
8  };
9  struct Node *adjList[MAX];
10 struct Node* createNode(int v)
11 {
12     struct Node *newNode;
13     newNode = (struct Node*)malloc(sizeof(struct Node));
14     newNode->vertex = v;
15     newNode->next = NULL;
16     return newNode;
17 }
18 void addEdge(int src, int dest)
19 {
20     struct Node *newNode;
21     newNode = createNode(dest);
22     newNode->next = adjList[src];
23     adjList[src] = newNode;
24     newNode = createNode(src);
25     newNode->next = adjList[dest];
26     adjList[dest] = newNode;
27 }
28 void display(int vertices)
29 {
30     int i;
31     struct Node *temp;
32     printf("\nAdjacency List:\n");
33     for(i = 0; i < vertices; i++)
34     {
35         printf("%d -> ", i);
```

C:\Users\Sevanthi-v\Downloads\DS lab 11 b.c C/C++ Windows (CR+LF) WINDOWS-1252 Line 66, Col 1, Pos 1460 Insert Read/Write default

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```
33     for(i = 0; i < vertices; i++)
34     {
35         printf("%d -> ", i);
36         temp = adjList[i];
37         while(temp != NULL)
38         {
39             printf("%d -> ", temp->vertex);
40             temp = temp->next;
41         }
42         printf("NULL\n");
43     }
44 }
45 int main()
46 {
47     int vertices, edges;
48     int i, src, dest;
49     printf("Enter number of vertices: ");
50     scanf("%d", &vertices);
51     for(i = 0; i < vertices; i++)
52     {
53         adjList[i] = NULL;
54     }
55     printf("Enter number of edges: ");
56     scanf("%d", &edges);
57     for(i = 0; i < edges; i++)
58     {
59         printf("Enter edge (source destination): ");
60         scanf("%d %d", &src, &dest);
61         addEdge(src, dest);
62     }
63     display(vertices);
64     return 0;
65 }
66
```

C:\Users\Sevanthi-v\Downloads\DS lab 11 b.cC/C++Windows (CR+LF)WINDOWS-1252Line 66, Col 1, Pos 1460InsertRead/Writedefault

```
Enter number of vertices: 1
Enter number of edges: 2
Enter edge (source destination): 89
234
Enter edge (source destination): 98
987

Process returned -1073741819 (0xC0000005)   execution time : 19.436 s
Press any key to continue.
```